



Answer the following questions.

Question 1: MCQ

1. Which of the following best distinguishes lemmatization from stemming in NLP?
☒ A. Lemmatization uses predefined suffix rules, while stemming depends on vocabulary lookup.
☐ B. Stemming uses context-based transformations, while lemmatization simply truncates words.
☒ C. Lemmatization considers the grammatical context and morphological analysis, while stemming does not.
☐ D. Stemming always produces valid dictionary words, while lemmatization may not.
2. Why is the Continuous Bag-of-Words (CBOW) model considered faster but less effective for rare words compared to Skip-gram?
☒ A. CBOW uses the target word to predict its surrounding context words.
☒ B. CBOW averages multiple context words to predict one target, reducing computational complexity but losing rare word sensitivity.
☐ C. CBOW uses a higher-dimensional hidden layer that overfits frequent words.
☐ D. CBOW requires large windows for rare words, increasing noise.
3. In Word2Vec training, what happens when the window size parameter is increased?
☒ A. It decreases the number of training samples generated.
☒ B. It increases the number of context words considered, thus capturing broader semantic relationships.
☐ C. It reduces overfitting by limiting context.
☐ D. It prevents the model from learning syntactic relations.
4. What is a fundamental computational drawback of Bag-of-Words (BoW) and One-Hot Encoding that Word Embeddings overcome?
☒ A. They cannot handle non-English vocabularies.
☒ B. They introduce high-dimensional sparse vectors that ignore semantic relationships between words.
☐ C. They require labeled data for training word representations. They rely on deep neural networks for vectorization.
5. In a Recurrent Neural Network (RNN), what is the primary role of the feedback loop?
☒ a) To propagate data forward only
☒ b) To enable the network to remember previous inputs
☐ c) To increase the number of output neurons
☐ d) To reduce the number of input connections
6. When unrolling an RNN through time, what is a key effect on the network structure?
☒ a) It converts the RNN into a feedforward network with multiple inputs and outputs
☐ b) It removes the biases from the network
☐ c) It consolidates multiple layers into one layer
☐ d) It eliminates the need for activation functions
7. Which of the following best describes the purpose of Word Embeddings in NLP?
☒ a) To represent text using word frequency counts only
☒ b) To convert words into numeric vectors that capture semantic and syntactic meaning
☐ c) To create sparse matrices for text classification
☒ d) To store each word as a unique binary index
8. What is a major limitation of the Bag-of-Words (BoW) model?
☒ a) It captures the semantic relationships between words
☐ b) It requires labeled data for training
☒ c) It ignores the order and context of words in a document
☐ d) It reduces dimensionality automatically
9. Which of the following best describes the difference between RNNs and standard feedforward neural networks?
☒ a) RNNs include feedback loops that enable temporal dynamic behavior
☐ b) RNNs do not use activation functions
☐ c) Feedforward networks are unrolled across time unlike RNNs
☐ d) RNNs do not have weights and biases

26. In the skip-gram model, what serves as the input and what is predicted as output?
 a) Input: context words, Output: target word b) Input: document, Output: topic c) Input: target word, Output: context words d) Input: frequency count, Output: embedding vector
27. What is the main purpose of the attention mechanism in sequence-to-sequence models?
 A. Reduce model size B. Speed up training C. Focus on relevant parts of the input sequence D. Eliminate the encoder
28. In Luong attention, what does the encoder produce for each input word?
 A. Output probabilities B. Hidden states C. Loss values D. Attention weights
29. How are attention scores computed in basic Luong attention?
 A. Cosine similarity B. Euclidean distance C. Dot product between decoder state and encoder states D. Manhattan distance
30. Why is the softmax function applied to attention scores?
 A. To remove negative values B. To normalize scores into probabilities C. To reduce computation D. To encode word positions
31. What does the context vector represent?
 A. Average of input embeddings B. Sum of decoder outputs C. Weighted sum of encoder hidden states D. Final translated word
32. During translation of "J'aime", which English word receives the highest attention?
 A. I B. love C. NLP D. All equally
33. What happens to attention weights when the decoder moves to the next word?
 A. They stay fixed B. They are randomly generated C. They are recalculated D. They are ignored
34. Which component decides how much importance each input word gets?
 A. Encoder B. Decoder C. Attention mechanism D. Softmax layer only
35. What advantage does attention provide over basic encoder-decoder models?
 A. Removes the need for training data B. Allows dynamic focus on different input words C. Eliminates vanishing gradients completely D. Works only for short sentences

Question 2: true or false

- The attention mechanism assigns the same importance to all input words.
- In Luong attention, attention scores are computed using the decoder hidden state.
- The context vector changes at each decoding step.
- Softmax converts attention scores into values that sum to 1.
- Attention mechanisms help improve translation quality for long sentences.

Question 3: essay

- Explain the Luong Attention mechanism in sequence-to-sequence models. [3 Marks]
- Describe step-by-step how the Luong Attention mechanism translates the sentence "I love NLP" into "J'aime le NLP". [5 Marks]
- Discuss the advantages of using an attention mechanism, such as Luong Attention, compared to a basic encoder-decoder model without attention. [3 Marks]
- Why were LSTM networks introduced as an extension of RNNs? [4 Marks]
- How does the forget gate in an LSTM determine what information to retain or discard? [4 Marks]

Question 4: Problem [7 Marks]

You have four data points as shown in the figure. The values of these four points can be observed from the graph.

Using the given weights, calculate the value of the fifth point using RELU activation function with its weight 1.2

and $w_1 = w_2 = w_3 = 1.9$ $w_4 = 1.1$ $w_5 = 1.3$

all bias = 1

